

3-2 Milestone Two: Enhancement One: Software Design and Engineering

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In this artifact, I implement a large amount of boilerplate structure to hoist up a modern Jetpack Compose Android application. I am implementing it in Kotlin. This has proven to be a large and complex task. My original Android application was written in Java and used imperative UI programming, where references to individual UI controls are obtained and their state, such as text labels and colors, are mutated directly. The Compose framework is one of declarative programming, where the UI is actually a function that takes a state and generates a resulting UI output. The entire UI function tree is “recomposed” smartly upon any state change.

For navigation between screens and maintenance of the backstack, I am using Jetpack Compose Navigation, which has also proven to be a difficult framework for me to master. I am currently still implementing the connective components necessary to wire up my Compose functions, ViewModels, and Navigation controller.

I am learning a lot about dependency injection and inversion of control, and the utility of dependency frameworks like Hilt (which I am not using at the moment). I am orienting myself away from the Model-View-Controller mindset to the Model-View-ViewModel framework which has many distinct differences.

I think I have not quite completed my outcome of implementing the entire “watering a garden” screen workflow, though I am getting close to this. It should be implemented by the end of the next module.

AI Assistance Used

While reading and trying to understand the documentation for Compose, NavController, ViewModels, and the dependency injection scheme in Android. I used Google Gemini, Google

Gemma3, OpenAI GPT-OSS:20b, Qwen3 for asking clarifying questions. I also used it to refine my understanding by giving my own explanation of these concepts and asking it to critique my explanations. I did not directly incorporate any code samples from any GenAI output into my application.

References